

EE 134 — Homework 1

Kaushik Vada

Problem 1.1

Given:

- NMOS: $V_{GS} = 1.8 \text{ V}$, $V_{th,n} = 0.7 \text{ V}$, $\mu_n C_{ox} = 200 \mu\text{A}/\text{V}^2$, $W/L = 10$.
- PMOS: $V_{SG} = 1.8 \text{ V}$, $|V_{th,p}| = 0.9 \text{ V}$, $\mu_p C_{ox} = 80 \mu\text{A}/\text{V}^2$, $W/L = 20$.

Assume both devices are in saturation and ignore channel-length modulation.

Model (saturation):

$$I_D = \frac{1}{2} \mu C_{ox} \frac{W}{L} (V_{OV})^2, \quad V_{OV} = V_{GS} - V_{th} \text{ (NMOS)}, \quad V_{OV} = V_{SG} - |V_{th,p}| \text{ (PMOS)}.$$

NMOS:

$$V_{OV,n} = V_{GS} - V_{th,n} = 1.8 - 0.7 = 1.1 \text{ V}$$

$$I_{D,n} = \frac{1}{2} (200 \mu\text{A}/\text{V}^2) (10) (1.1)^2 = \frac{1}{2} (2000 \mu\text{A}/\text{V}^2) (1.21) = 1210 \mu\text{A} = 1.21 \text{ mA}.$$

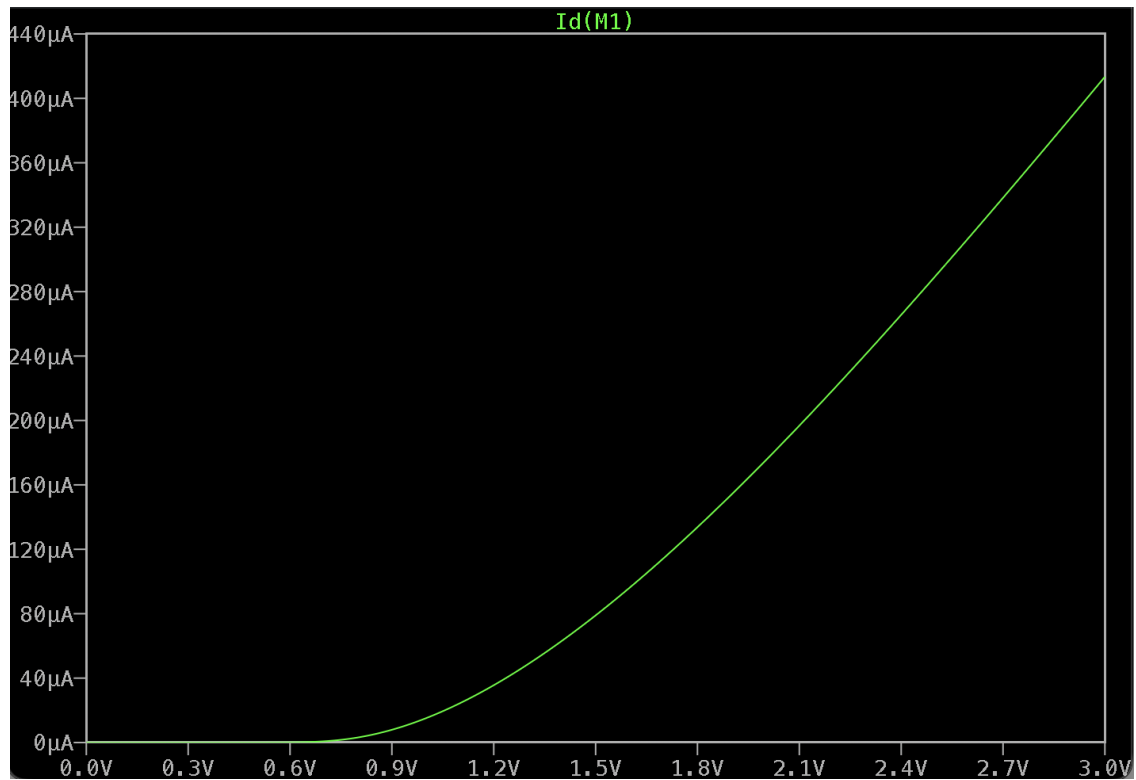
PMOS:

$$V_{OV,p} = V_{SG} - |V_{th,p}| = 1.8 - 0.9 = 0.9 \text{ V}$$

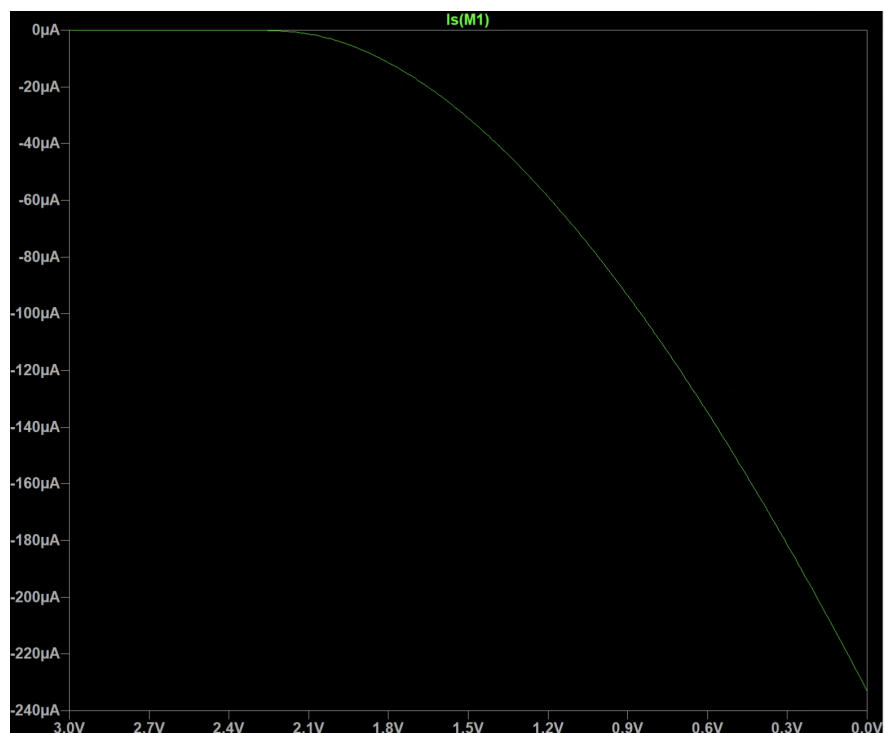
$$I_{D,p} = \frac{1}{2} (80 \mu\text{A}/\text{V}^2) (20) (0.9)^2 = \frac{1}{2} (1600 \mu\text{A}/\text{V}^2) (0.81) = 648 \mu\text{A} = 0.648 \text{ mA}.$$

Why PMOS are often wider in digital cells: PMOS devices are usually weaker because holes move slower than electrons ($\mu_p < \mu_n$). So designers make the PMOS wider (bigger W/L) to increase its drive current and make the rise time closer to the fall time.

Problem 1.2



Problem 1.3



Problem 1.4(a) and 1.4(b)

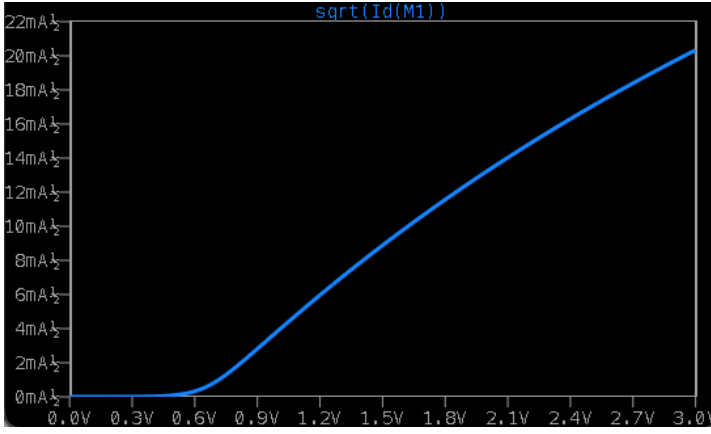


Figure 1: *
(a)

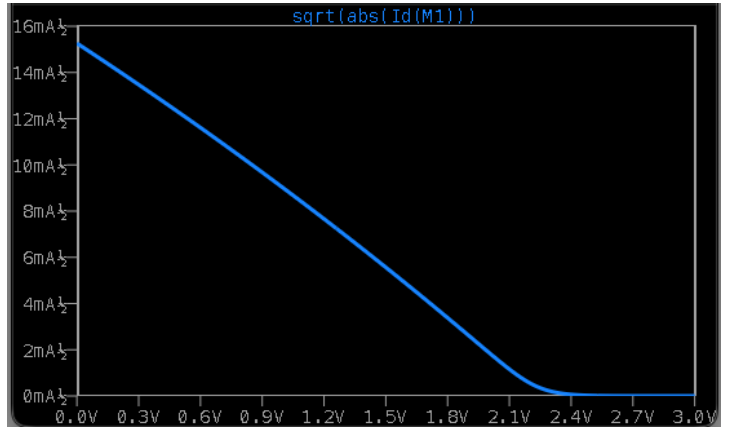
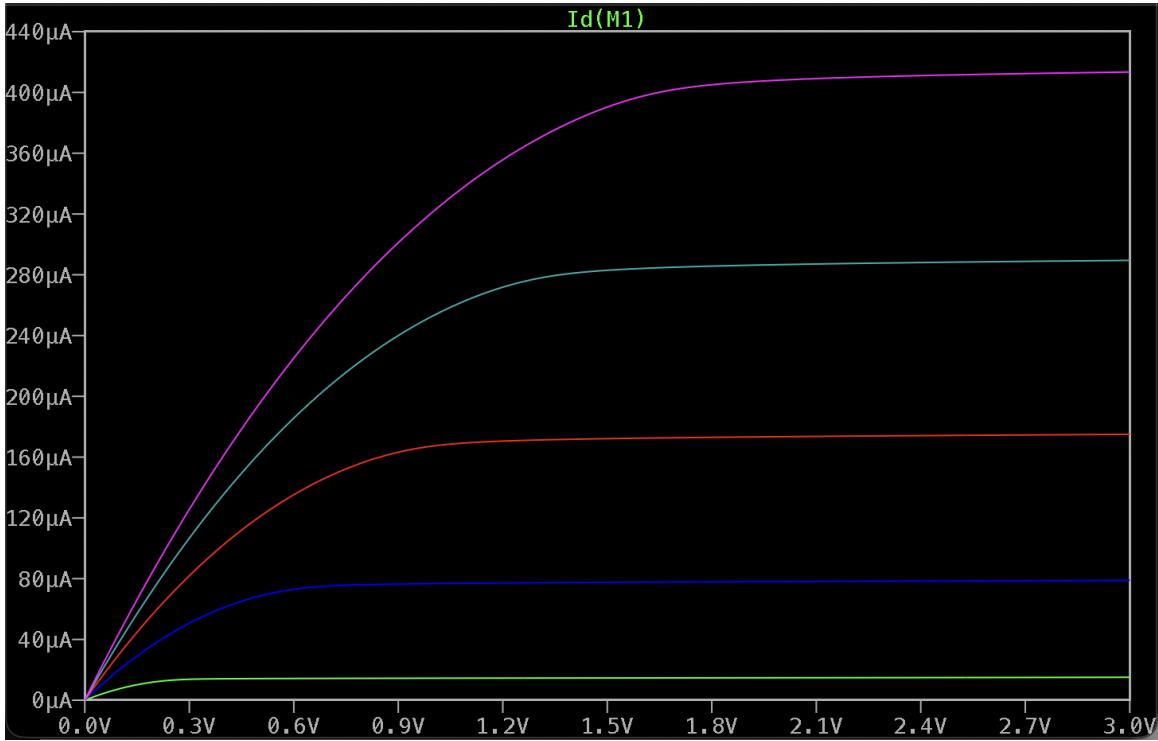


Figure 2: *
(b)

By extrapolating the linear region of the $\sqrt{I_D}$ vs. V_G curve, the intercept was found to be approximately 0.712 V, which is the estimated threshold voltage.

By extrapolating the linear region of the $\sqrt{|I_D|}$ vs. V_G curve, the intercept was found to be approximately 2.15 V. Since the PMOS source is at 3.0 V, the estimated threshold voltage is $|V_{th}| = 3.0 - 2.15 = \mathbf{0.85 \text{ V}}$.

Problem 1.5

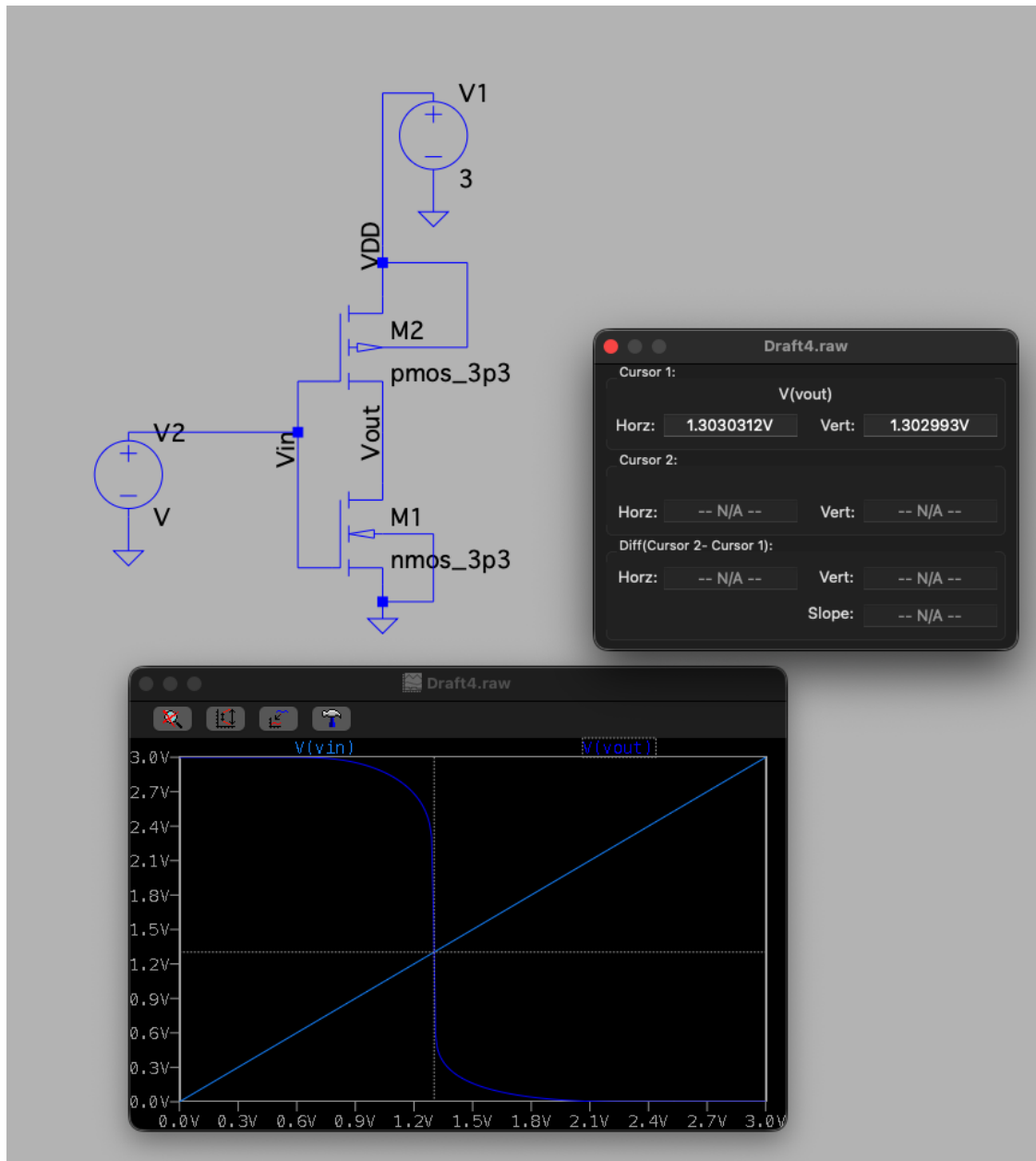


Transition Points:

- $V_{GS} = 1.0 \text{ V}$: Transition $\approx 0.29 \text{ V}$
- $V_{GS} = 1.5 \text{ V}$: Transition $\approx 0.79 \text{ V}$
- $V_{GS} = 2.0 \text{ V}$: Transition $\approx 1.29 \text{ V}$
- $V_{GS} = 2.5 \text{ V}$: Transition $\approx 1.79 \text{ V}$
- $V_{GS} = 3.0 \text{ V}$: Transition $\approx 2.29 \text{ V}$

Method: I approximated the transitions by identifying the “knee” where the current slope flattens. These points align with the theoretical saturation condition $V_{DS} \geq V_{GS} - V_{th}$, using the $V_{th} \approx 0.71 \text{ V}$ estimated in Q1.4.

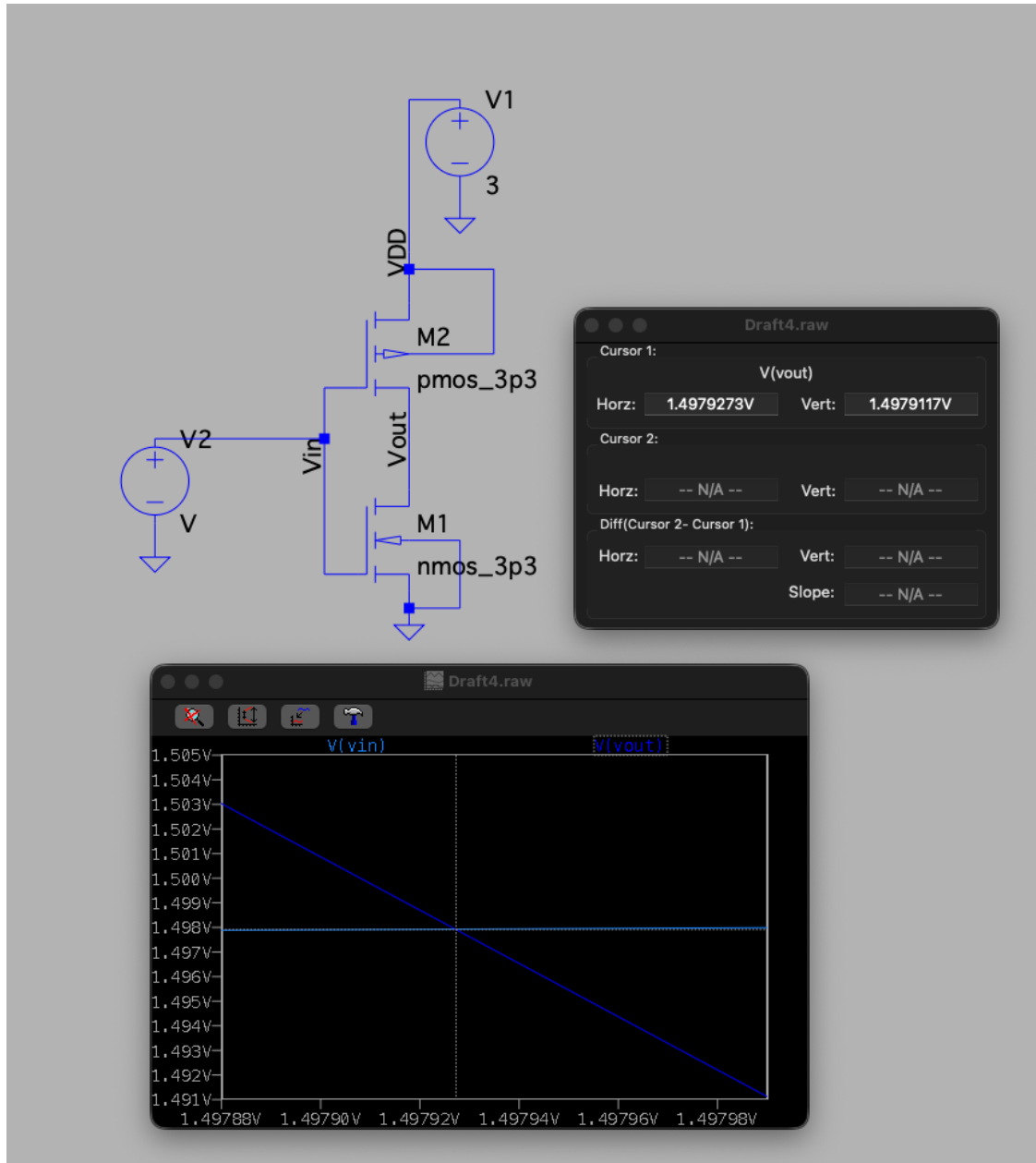
Problem 1.6



Answer.

- **Switching threshold:** The intersection point (switching threshold) is $V_M = 1.30\text{ V}$.
- **Compared to ideal:** This is lower than the ideal midpoint $V_{DD}/2 = 1.5\text{ V}$.
- **Why it shifts:** The NMOS is effectively *stronger* than the PMOS. Even though the PMOS width ($4\text{ }\mu\text{m}$) is double the NMOS width ($2\text{ }\mu\text{m}$), it still doesn't fully compensate for the mobility difference.

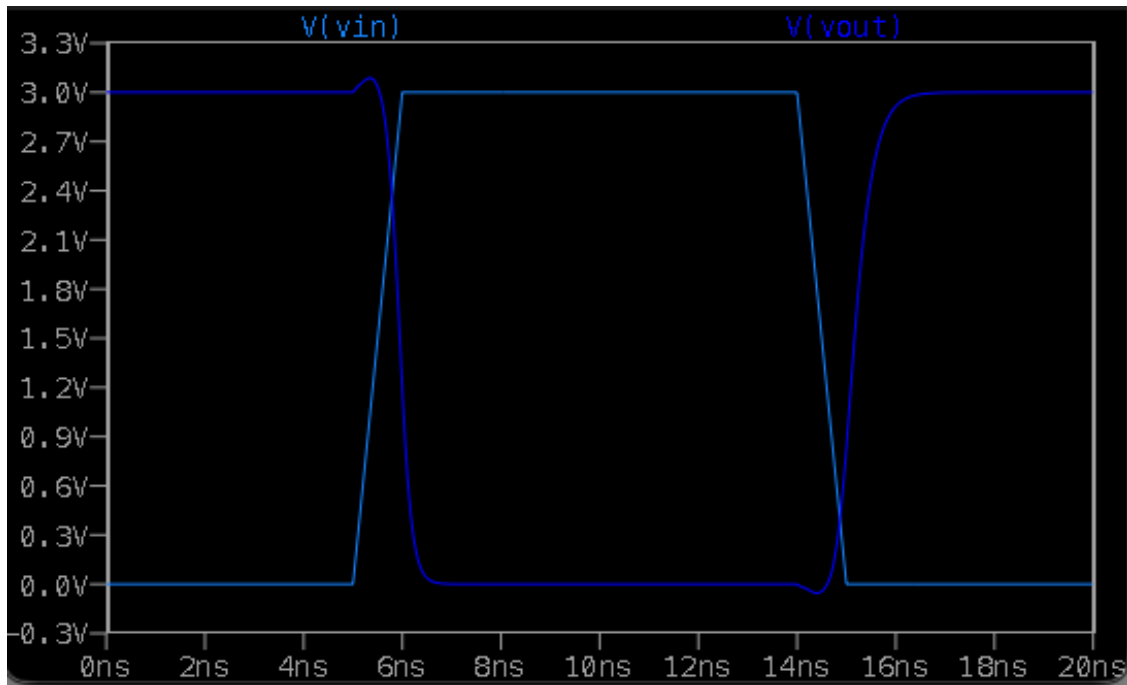
Problem 1.7



Answer.

- **Method:** To shift the switching threshold (V_M) to 1.5 V (half V_{DD}), I increased the width of the PMOS transistor to increase its current drive capability and better match the NMOS strength.
- **Results:**
 - **New PMOS width:** $W_p = 10 \mu\text{m}$ (NMOS kept at $2 \mu\text{m}$).
 - **Intersection:** The simulation shows the input and output voltages intersecting at approximately 1.498 V, which is within 0.2% of the 1.5 V target.

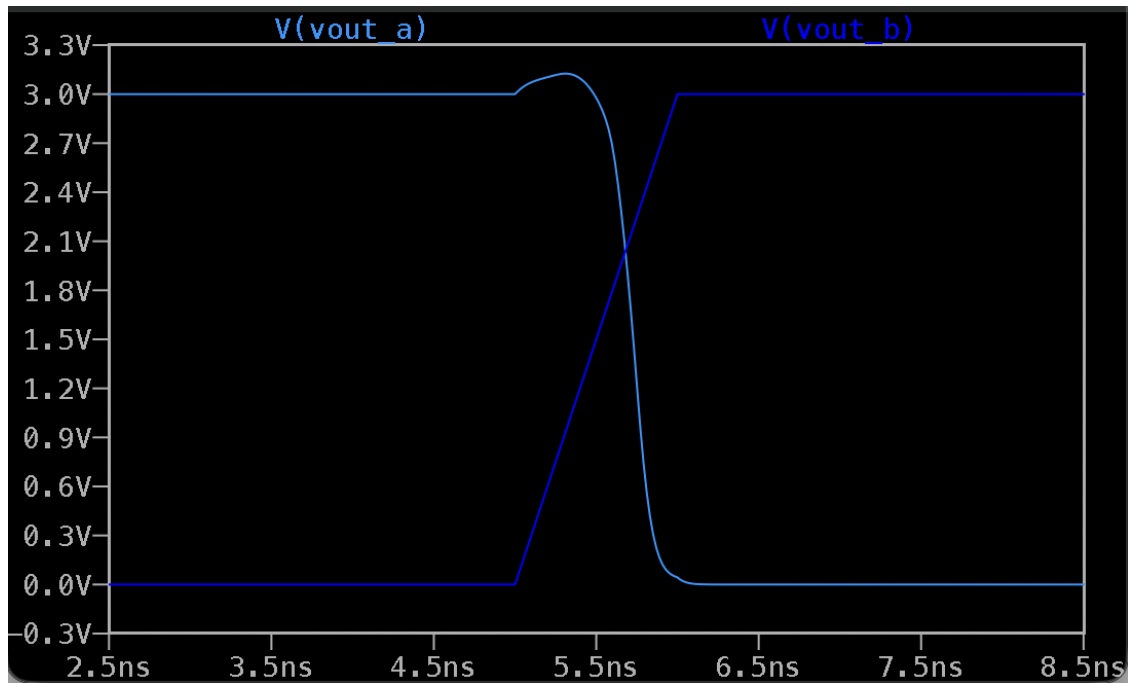
Problem 1.8



Answer.

- **Method:** I performed a transient simulation (.tran) and measured the time difference between the 50% voltage points (1.5 V) of the input and output waveforms.
- **Results:**
 - **High-to-Low delay** (t_{pHL}): 0.40 ns
 - **Low-to-High delay** (t_{pLH}): 0.67 ns
 - **Average propagation delay** (t_p): 0.535 ns
- **Observation:** The low-to-high delay (t_{pLH}) is longer than the high-to-low delay (t_{pHL}). This is expected because the PMOS (which pulls the output high) has lower mobility than the NMOS, so it charges the load capacitor more slowly even if it is wider.

Problem 1.9



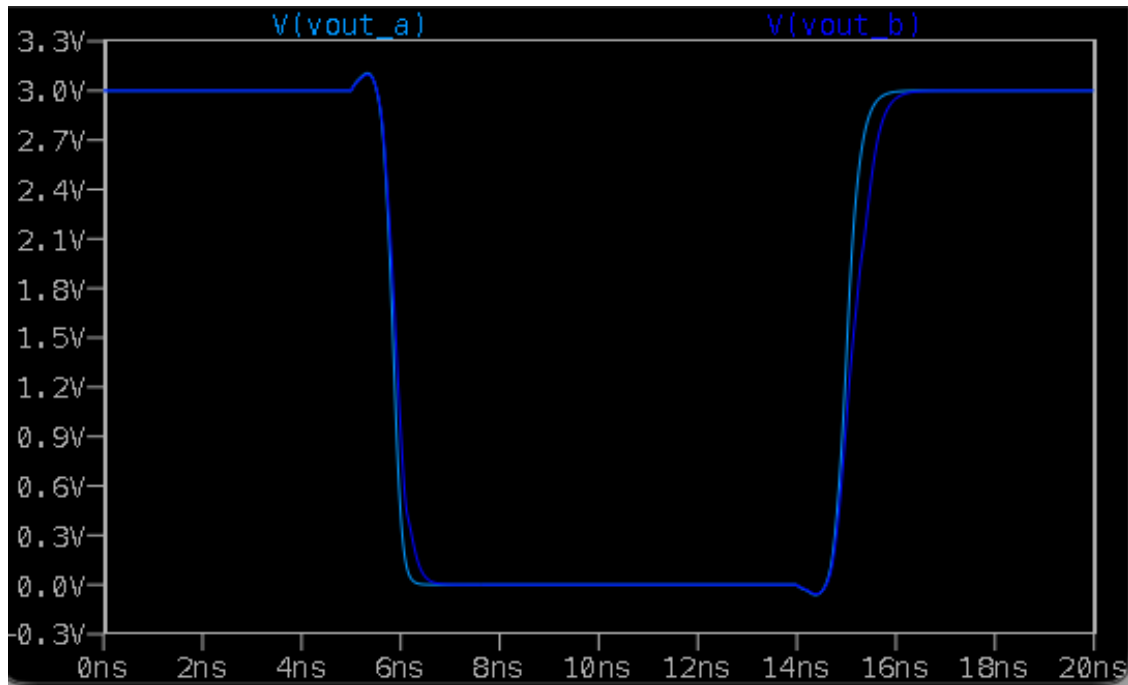
Answer.

- **Method:** To estimate the input capacitance (C_{in}) of the inverter, I set up a “test bench” with two parallel inverter chains.
 - **Chain 1 (Reference):** An inverter driving a discrete capacitor load (C_{load}).
 - **Chain 2 (Test):** An inverter driving the gate of a second inverter (Inverter C).

I tuned the value of C_{load} until the transient delay of the Reference inverter perfectly matched the delay of the Test inverter.

- **Result:** As shown in the waveform (Figure Q1.9), the output curves overlap perfectly when $C_{load} = 10$ fF. Therefore, the estimated input capacitance of the inverter is **10 fF**.

Problem 1.10



Answer.

- **Method:** I repeated the estimation procedure from Q1.9, but changed the dimensions of the test load (Inverter C) to NMOS ($4\mu\text{m}/1\mu\text{m}$) and PMOS ($8\mu\text{m}/1\mu\text{m}$). This effectively doubles the width (W) of the transistors compared to the inverter used in Q1.9.
- **Results:**
 - **Matched delay:** The reference inverter's delay matched the test chain when the load capacitor was set to approximately $C_{load} \approx 47\text{fF}$ (see Figure Q1.10).
- **Explanation:** The estimated input capacitance for Inverter C is significantly larger than the value found in Q1.9 (10fF). This increase occurs because the input capacitance is directly proportional to the physical gate area of the transistors ($C_{gate} \approx C_{ox} \cdot W \cdot L$). By increasing W of the NMOS and PMOS devices, we increased the total gate area, resulting in a higher input capacitance that must be charged and discharged during switching.